ZapBox Compact – Operating Instructions

Language: English | **Version:** OI940117

Overview

The **ZapBox Compact** is an electronic switch for Bitcoin Lightning payments. A payment via the Lightning Network triggers an output — ideal for vending machines, presentations, event control, and many other applications.

Base Equipment

Component	Description
Microcontroller	T-Display-S3 with 1.9" LCD display
Front panel	Available with 35° or 90° display front (90° also available for panel mounting)
Input	USB-C (Power IN)
Output	USB-A socket
Controls	Two on-board micro buttons
Switch 1	3-position slide switch (AUTO / OFF / ON)
Switch 2	2-position slide switch (Invert Output)
Option BTC Ticker	Activatable via the Web Installer
Option NFC Module	For Bolt Cards (NTAG424 DNA) and standard NTAG213/215/216 (with LNURL-withdraw)

Views

Figure 1: Front view

Figure 2: Rear view

Getting Started

Power Supply

Supply the device via the **Power IN** connector using a USB-C cable with **5 V DC (max. 5 A)**.

> **Note:** The Power IN connector is not "intelligent". Some chargers or power modules with a USB-C output do not recognise the ZapBox and will not supply power. In this case, use a **USB-A output** of the power supply or a different power source.

First Test (pre-configured device)

If the device is already configured, you can immediately perform a first test:

1. Press **NEXT** to cycle through the display pages.
2. Scan the displayed **QR code** with a Lightning wallet.
3. Pay the invoice.
4. The relay should audibly switch — the switching operation was successful.

Setup

Web Installer

For initial setup and firmware updates, the **Web Installer** is available:

https://installer.zapbox.space/

The complete procedure is described there. It is recommended to always flash the **"Latest" firmware** to keep the ZapBox up to date.

USB Data Access (hidden cover)

To read or transfer data from the device, connect the ZapBox to a computer or laptop:

1. On the **right side of the front panel** there is a small, concealed cover.
2. Open the cover by sliding it to the right using a **narrow screwdriver** from below.
3. Connect a USB-C cable to the microcontroller connector underneath.

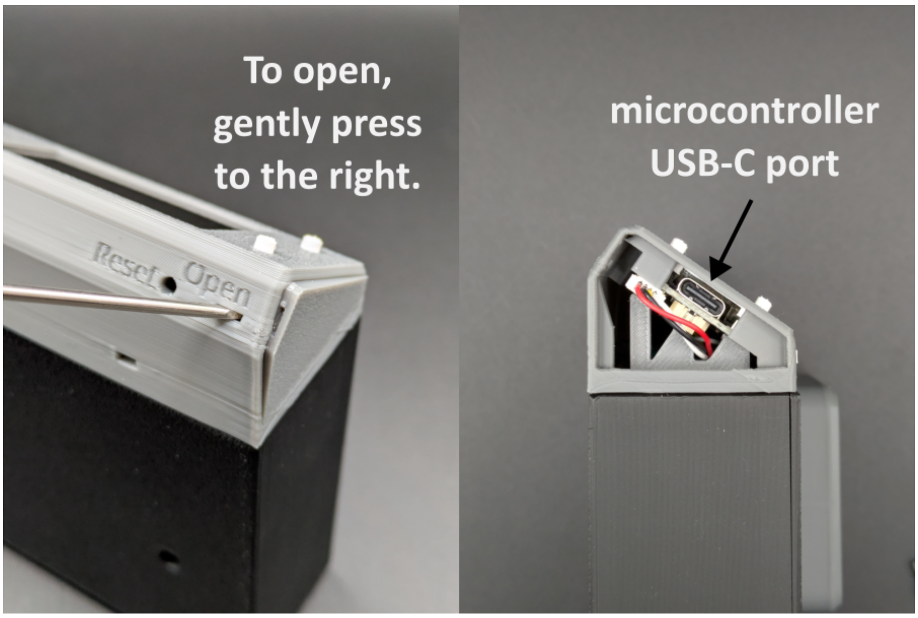

Figure: Opening the panel and USB-C connector for data

> **Important note:** The USB connector directly on the microcontroller is used exclusively for flashing new firmware or transferring configuration parameters. If switching functions are triggered simultaneously during flashing, **malfunctions or damage to the microcontroller** may occur.
>
> It is therefore recommended to:

> - Not trigger any switching functions during flashing, **or**
> - additionally connect the regular **Power IN input** to the same power supply, so that the current for the power relay does not flow through the microcontroller and overload it.

Slide Switches

The ZapBox Compact has two slide switches.

Switch 1 – Triple Slide Switch (AUTO / OFF / ON)

Position	Function
A (AUTO)	Automatic mode – normal operation
0 (OFF)	Power supply interrupted – output OFF
1 (ON)	Output permanently ON

Switch 2 – Dual Slide Switch (Invert / Normal)

Position	Function
Normal	The output is de-energised at rest (0 V). After switching, 5 V is present at the USB sockets.
Inv. (Invert)	The output is at 5 V at rest. After switching, the output changes to 0 V (inverted switching).

> **Note:** If the dual switch is set to **Inv.** and the triple switch is in **position 1**, the output is — unlike in normal operation — **OFF** instead of ON.

Output: USB-A Socket

The USB socket is switched via a relay contact. The **total load** on the sockets should **not exceed 3 A**.

NFC Module (optional)

Depending on the configuration, an NFC module is installed on the **top of the ZapBox**. It supports the following card types:

- **Bolt Cards** (NTAG424)
- **LNURL-Withdraw** from NTAG21x (213 / 215 / 216)

Controls

The ZapBox has two small **on-board micro buttons** connected directly to the microcontroller. All functions are accessible via the micro buttons. The ZapBox also has a Reset button on the underside.

Function Overview

Function	Micro Button
Show help page	Press HELP once
Next page / product change	Press NEXT once
Show REPORT page	Press HELP twice
Enter config mode	Hold NEXT for at least 5 seconds

Technical Specifications

Property	Value
Supply voltage	5 V DC via USB-C
Maximum input current	3.5 A
Output power	max. 3.0 A
Display	1.9" LCD (T-Display-S3)
Communication	Wi-Fi (ESP32-S3)
Payment protocol	Bitcoin Lightning Network

Safety Instructions

- Operate the device only with the specified supply voltage.
- Do not exceed the maximum current ratings of the outputs.
- Do not work on relay contacts under load.
- The device is not suitable for use in humid or wet environments.
- Keep out of reach of children.

Further Links

Resource	Link
Overview of all ZapBox models	https://zapbox.space/
Web Installer, quick overview & troubleshooting	https://installer.zapbox.space/
Detailed documentation (parameters & functions)	https://ereignishorizont.xyz/zapbox/
GitHub repository (software, PCB layouts, 3D print files)	https://github.com/AxelHamburch/ZapBox

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Subject to changes and errors. As of: 2026